# Flexible Attention-Based Multi-Policy Fusion for Efficient Deep Reinforcement Learning


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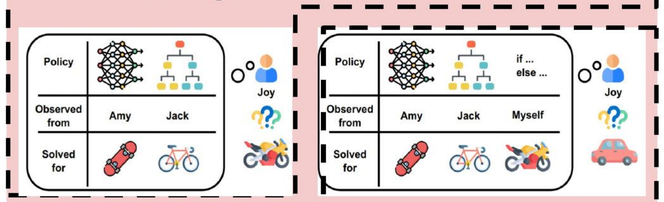
## Why Knowledge-Grounded Reinforcement Learning?

- What learning capabilities do humans possess, yet RL agents still missing? [1,2]
- **Knowledge-Acquirable**: Develop unique policies by observing.
- **Compositional**: Combine policies from multiple sources to form a knowledge set.

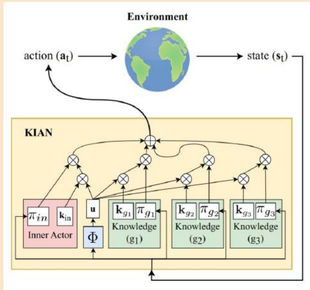



- **Incremental**: Arbitrarily remove outdated policies or add new ones, no need to relearn how to navigate the entire knowledge set from scratch.
- **Sample-Efficient**: Require fewer interactions.
- **Generalizable**: Be able to apply previously learned policies to unseen tasks.
- **KGRL**: An RL paradigm aiming to learn with external policies and achieve the five properties above.

## Method: Knowledge-Inclusive Attention Network (KIAN)

- **KIAN:** An actor that separates knowledge representations from their selection mechanism.

Model Architecture:
- Inner Actor
- Knowledge Keys
- Query



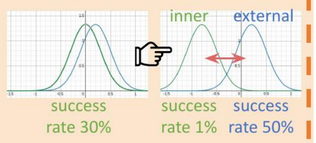

- Naive policy fusion leads to challenges:

$$\pi(\cdot|\mathbf{s}_t) = \hat{w}_{in}\pi_{in}(\cdot|\mathbf{s}_t) + \Sigma_{j=1}^{n}\hat{w}_{g_j}\pi_{g_j}(\cdot|\mathbf{s}_t)$$

- **Discrete vs Continuous** action space
- **Entropy Imbalance** in maximum entropy KGRL

**P1: always inner actor**

**P2: distant means**




### Discrete Action Policy

$$\pi(\cdot|\mathbf{s}_t) = \hat{w}_{t,in}\pi_{in}(\cdot|\mathbf{s}_t) + \Sigma_{j=1}^{n}\hat{w}_{t,g_j}\texttt{softmax}(\beta_{t,g_j}\pi_{g_j}(\cdot|\mathbf{s}_t))$$

$$w_{t,g_j} = \Phi(\mathbf{s}_t)\cdot\mathbf{k}_{g_j}/\beta_{t,g_j} \quad \beta_{t,g_j} = \|\Phi(\mathbf{s}_t)\|_2\|\mathbf{k}_{g_j}\|_2$$

### Continuous Action Prediction

$$\pi_{in}(\cdot|\mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_{t,in}, \boldsymbol{\sigma}^2_{t,in}), \pi_{g_j}(\cdot|\mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_{t,g_j}, \boldsymbol{\sigma}^2_{t,g_j})$$

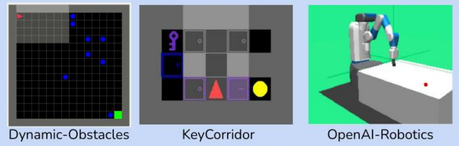
$$e \sim \texttt{gumbel_softmax}([\hat{w}_{t,in}, \hat{w}_{t,g_1}, \ldots, \hat{w}_{t,g_n}]^\top) \quad \mathbf{a}_t \sim \pi_e(\cdot|\mathbf{s}_t)$$

### Continuous Probability Distribution

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \hat{w}_{in}\pi_{in}(\mathbf{a}_{t,in}|\mathbf{s}_t) + \Sigma_{j=1}^{n}\hat{w}_{g_j}\pi_{g_j}(\boldsymbol{\mu}_{t,g_j}|\mathbf{s}_t)$$

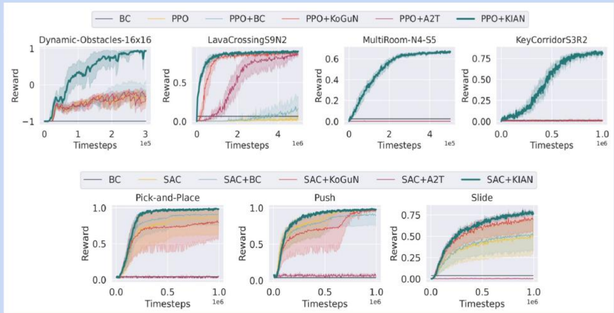
## How well are the KGRL properties satisfied?

- Environments:
  - Discrete decision making (MiniGrid)
  - Continuous control (OpenAI-Robotics)




- RL algorithms: proximal policy optimization (PPO) and soft actor critic (SAC)
- **Main Results:**

### Sample Efficiency & Compositional



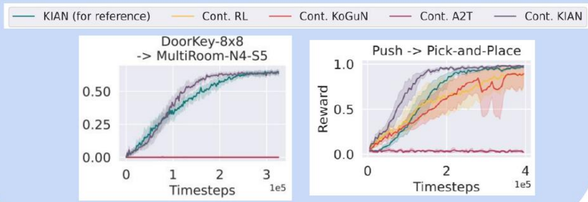

### Generalizability

- Zero-shot transfer policies from simple to more complex tasks

| Train in / Test in | Empty-Random-5x5 6x6 | 8x8 | 16x16 | DoorKey-5x5 8x8 | 16x16 | Push 5x | 10x | Slide 5x | 10x | Pick-and-Place 5x | 10x |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RL | 0.88 | 0.71 | 0.45 | 0.29 | 0.08 | 0.87 | 0.52 | 0.45 | 0.17 | 0.34 | 0.27 |
| RL+BC | 0.87 | 0.60 | 0.24 | 0.40 | 0.09 | 0.89 | 0.60 | 0.44 | 0.16 | 0.34 | 0.30 |
| KoGuN [3] | 0.94 | 0.83 | 0.53 | **0.77** | 0.35 | 0.63 | 0.43 | **0.55** | **0.18** | 0.32 | 0.24 |
| A2T [4] | 0.92 | 0.78 | 0.51 | 0.53 | 0.11 | 0.03 | 0.05 | 0.00 | 0.01 | 0.01 | 0.06 |
| KIAN (ours) | **0.96** | **0.91** | **0.93** | 0.76 | **0.42** | **0.93** | **0.70** | 0.42 | 0.15 | **0.92** | **0.72** |

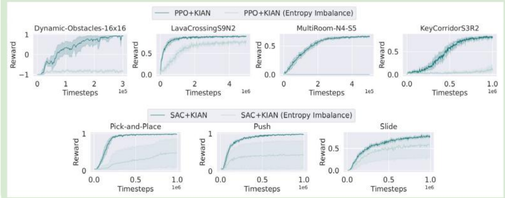
### Compositional & Incremental

- Combine previously learned knowledge keys and inner policies to learn a new task




## How entropy imbalance affects?

- Addressing entropy imbalance is important!

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